



Scope of Work (SoW)

Manufacturing ERP System

Date: January 9, 2026

Client: Manufactuer

Estimated Duration: ~3 Weeks

Prepared by: GBG Academy Students

Mohamed Samir

Ziad Mohamed

Abdelrahman Adam



1. Executive Summary

This engagement delivers a secure, production-oriented Azure foundation and migrates an ERP workload to Azure with an Active/Passive DR model. The solution uses Hub-and-Spoke networking, defense-in-depth security, Azure Front Door (with WAF) for global access, and Azure Site Recovery (ASR) for orchestration and failover. The ERP workload consists of two VMs (SQL + IIS Web). Hybrid management is demonstrated via Azure Arc for an on-prem SQL instance.

Category	Deliverable
Network Architecture	Hub-Spoke topology with 3 spokes (Database, Application, Storage)
Compute	2 Virtual Machines (SQL Server, IIS Web Server)
Security	Azure Firewall, WAF, NSGs, Private Endpoints
Connectivity	Point-to-Site VPN for secure remote access
DR Solution	Azure Site Recovery with automatic failover
Global Access	Azure Front Door with intelligent load balancing
Hybrid Cloud	Azure Arc-enabled SQL Server on-premises



2. Objectives and Success Criteria

Objectives

- Migrate ERP workload from on-prem to Azure
- Implement **Active/Passive DR** with **RPO < 15 minutes** and **RTO < 1 hour**
- Enforce network segmentation (Hub-Spoke), NSGs, and centralized firewall controls
- Provide secure admin access via **Point-to-Site VPN**
- Enable hybrid visibility/management via **Azure Arc**

Success Criteria (High-level)

- Availability target: **99.9%** (architecture + DR readiness)
- ASR replication healthy and meeting RPO targets
- failover procedures validated (test failover evidence)
- Security segmentation enforced (hub-spoke isolation)
- Cost tracked via Azure Cost Management against target



3. In-Scope Work

3.1 Network Architecture (Hub-Spoke)

- Build Hub VNet (Italy North) with:
 - **Azure Firewall (Standard)**
 - **VPN Gateway (P2S)**
- Build 3 spokes:
 - **DB Spoke** (SQL VM)
 - **App Spoke** (IIS Web VM)
 - **Storage Spoke** (Azure Files / Blob + Private Endpoints)
- Configure VNet peering (hub-to-spokes), and routing (UDRs) to force traffic via firewall where required.

3.2 Security Baseline

- Azure Firewall (primary + DR) and firewall policy alignment
- Front Door WAF
- NSGs per spoke/subnet
- Private Endpoints for Storage + Private DNS Zones (Files/Blob)
- “No public IPs on VMs” design assumption (private access via VPN + Front Door).



3.3 Compute (ERP Workload)

- Deploy 2 VMs:
 - SQL Server VM (spec defined in file; VM sizing based on cost-optimized SKU)
 - IIS Web Server VM (IIS + Python/Flask per demo design)

3.4 Storage

- Storage account for **Azure Files** and **Blob**
- Private Endpoints for file/blob access
- Private DNS Zones linked to VNets for name resolution.

3.5 Global Access Layer

- Azure Front Door (Standard) configured for:
 - WAF policy
 - Origin group with priority-based failover (Primary/DR).

3.6 Disaster Recovery (DR)

- Recovery Services Vault in **North Europe**
- Enable ASR replication for both VMs
- Create DR hub-spoke network in DR region (firewall, routing, peerings)
- Configure/validate Front Door failover behavior to DR origin
- Execute DR test failover and document results.



3.7 Hybrid Cloud (Demo)

- Azure Arc-enabled server (on-prem / Hyper-V VM)
- Azure Arc SQL extension for a demo SQL instance.

4. Out of Scope

- ERP application code changes, refactoring, or modernization.
- Azure Bastion, ExpressRoute (VPN P2S used instead)
- Production data migration (demo/sample data only).
- Custom domain + SSL certificates (Azure-provided domains used)
- Licensing procurement (Windows/SQL/third-party) unless explicitly assigned.
- End-user device configuration beyond providing VPN client packages and instructions.
- Multi-subscription enterprise landing zone hierarchy (single subscription)



5. Deliverables

#	DELIVERABLE	DESCRIPTION
1	SOW	project definition
2	DR Runbook documentation	failover/failback steps + validation checklist
3	Sizing for SQL VMs, storage, and all components	VM sizing + storage + SKU recommendations
4	Hub & Spoke Architecture Design	High-level and low-level design for 1 Hub + 3 Spokes (App/DB/Storage), connectivity flows, security controls, routing approach, and service placement.
5	Subnet Segmentation	CIDRs, subnet purpose, NSG boundaries
6	Phased Project Plan	milestones, dependencies, timeline

6. Acceptance Criteria

The project will be considered complete when:

- VPN connectivity works; admins can access VMs via private IPs
- Web VM connects to SQL VM on required ports (e.g., 1433)
- ERP application reachable via Front Door endpoint
- Storage private endpoint resolution and access works



- Security posture validated (no direct public VM access, logging, WAF behavior)
- DR replication healthy; test failover succeeds; Front Door failover validated; failback possible.

7. High-Level Implementation Phases

1. Foundation (RGs, naming, tags)
2. Hub network build
3. Spoke networks + NSGs
4. VNet peering + routing
5. VPN Gateway setup (certs + P2S)
6. Deploy SQL VM + Web VM
7. Storage + Private Endpoints + Private DNS
8. Front Door + WAF setup
9. ASR replication + DR hub build + DR firewall
10. Azure Arc + Arc SQL extension
11. Documentation + testing + DR drill(s) + final review/handover

8. Estimated Costs (High-Level)

Daily estimate (order-of-magnitude): ~\$22/day for the full demo stack (VMs + firewalls + VPN + Front Door + storage + ASR).